# Essential Summary

## A Generalized Padé–Lindstedt–Poincaré Method for Predicting Homoclinic and Heteroclinic Bifurcations of Strongly Nonlinear Autonomous Oscillators

Zhenbo Li, Jiashi Tang

*Nonlinear Dynamics, Vol. 84, No. 3, 1201–1223 (2016) DOI: 10.1007/s11071-015-2563-6*



# 1 Background and Motivation

The Lindstedt–Poincaré (L–P) method is a classical perturbation technique for weakly nonlinear oscillators. However, for strongly nonlinear systems—and especially for predicting **global bifurcations** (homoclinic/heteroclinic)—the traditional L–P method faces two fundamental obstacles: (1) the perturbation procedure requires repeated derivation and integration, which becomes prohibitively cumbersome at high orders for complicated oscillators; and (2) the method was originally designed for periodic solutions, not for the limit-cycle-to-homoclinic transition regime.

This work introduces the **generalized Padé–Lindstedt–Poincaré (GPLP) method**—a synthesis that replaces the integration steps in the L–P procedure with generalized Padé approximation, thereby eliminating the derivation/integration bottleneck while retaining the systematic perturbation framework. This is the foundational paper of the GPLP method, later extended by the authors to more complex oscillators (J. Vib. Eng. Technol., 2022).

# 2 Method: GPLP Framework

## 2.1 The Core Idea

Traditional L–P method steps: (1) introduce nonlinear time transformation $\varphi = \omega t$, (2) expand solution and frequency in powers of $\varepsilon$, (3) solve each-order perturbation equation by **integration** to eliminate secular terms. Step (3) is the bottleneck—for oscillators with complicated restoring forces, the integrals become intractable.

GPLP replaces step (3) with a **generalized Padé approximation** procedure:

1. Assume the solution to each perturbation order as a generalized Padé approximant (rational function of harmonic functions)
2. Determine its coefficients by matching with the perturbation equation—**no integration required**
3. The frequency $\omega$ is obtained simultaneously from the matching conditions

## 2.2 Generalized Padé Approximants Employed

Several types of approximants are constructed and tested:

- **Symmetric homoclinic:** $x(\varphi) = \frac{\sum_{i=0}^{m} \alpha_i \cos^{2i}\varphi}{\sum_{j=0}^{n} \beta_j \cos^{2j}\varphi}$ — exploits symmetry of homoclinic orbits
- **Asymmetric heteroclinic:** $x(\varphi) = \frac{\sum_{i=0}^{m} \alpha_i \sin^{2i}\varphi}{\sum_{j=0}^{n} \beta_j \sin^{2j}\varphi}$ — accommodates asymmetric saddle connections
- **Mixed type:** Combinations for oscillators where homoclinic and heteroclinic orbits coexist

### 2.3 Advantages Over Traditional L–P

1. **No derivation/integration:** The most time-consuming steps are eliminated. High-order solutions become accessible.
2. **Handles strong nonlinearity:** Since Padé approximants are global rational functions (not local Taylor expansions), they capture strongly nonlinear behavior well.
3. **Direct bifurcation prediction:** By treating the bifurcation parameter analytically, the critical homoclinic/heteroclinic values are obtained directly from the amplitude–parameter relation.

## 3 Applications

### 3.1 Helmholtz–Duffing Oscillator

The asymmetric Helmholtz–Duffing oscillator serves as a benchmark for heteroclinic bifurcation prediction. GPLP provides analytical homoclinic/heteroclinic solutions and predicts critical bifurcation parameters (Fig. 1).

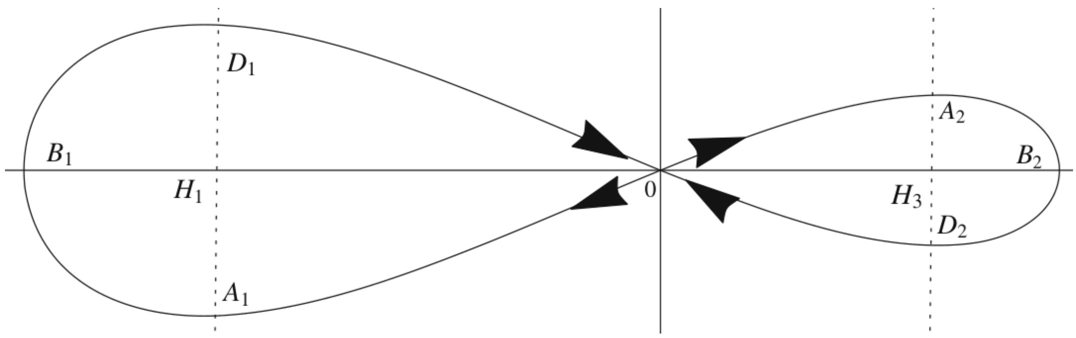


Figure 1: Potential energy curve and phase portraits for the Helmholtz–Duffing oscillator.

### 3.2 Duffing-Harmonic Oscillator

For oscillators with rational restoring forces, GPLP handles homoclinic orbits under both small ($\varepsilon = 0.05$) and moderate ($\varepsilon = 0.1$) perturbation parameters (Fig. 3).

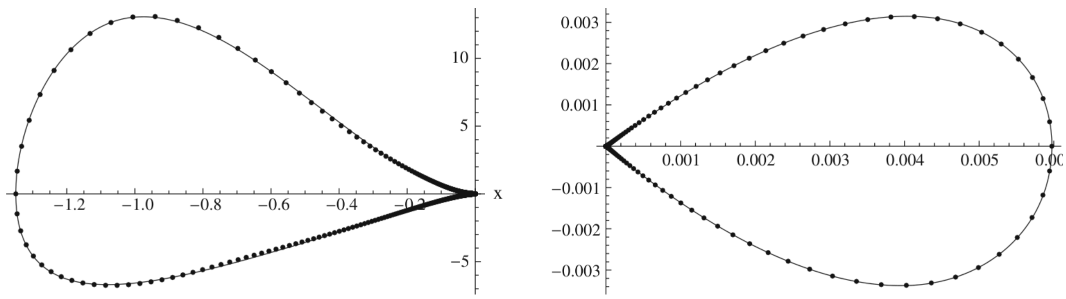


Figure 2: Homoclinic orbits of the Duffing-harmonic oscillator for $\varepsilon = 0.05$ (a) and $\varepsilon = 0.1$ (b). Solid: Runge–Kutta. Dashed: GPLP.

### 3.3 SD Oscillator (Multi-Well)

The irrational SD oscillator is the most challenging test case. GPLP successfully predicts homoclinic orbits in both double-well and triple-well configurations (Figs. 4–5).

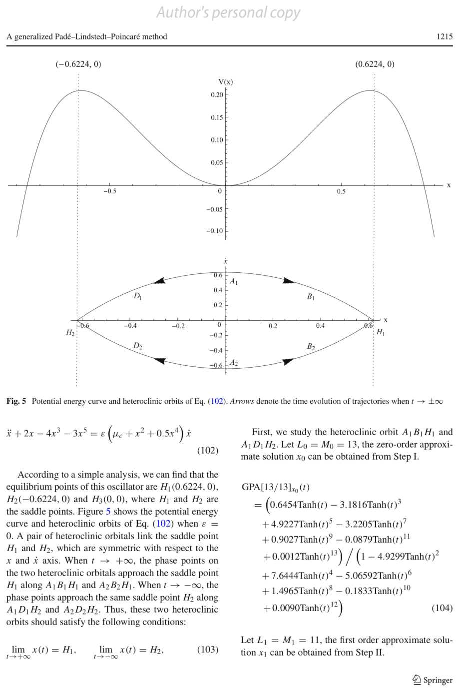



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**Fig. 5** Potential energy curve and heteroclinic orbits of Eq. (102). *Arrows* denote the time evolution of trajectories when $t \to \pm\infty$

$$\ddot{x} + 2x - 4x^3 - 3x^5 = \varepsilon\left(\mu_c + x^2 + 0.5x^4\right)\dot{x} \tag{102}$$

According to a simple analysis, we can find that the equilibrium points of this oscillator are $H_1(0.6224, 0)$, $H_2(-0.6224, 0)$ and $H_3(0, 0)$, where $H_1$ and $H_2$ are the saddle points. Figure 5 shows the potential energy curve and heteroclinic orbits of Eq. (102) when $\varepsilon = 0$. A pair of heteroclinic orbitals link the saddle point $H_1$ and $H_2$, which are symmetric with respect to the $x$ and $\dot{x}$ axis. When $t \to +\infty$, the phase points on the two heteroclinic orbitals approach the saddle point $H_1$ along $A_1B_1H_1$ and $A_2B_2H_1$. When $t \to -\infty$, the phase points approach the same saddle point $H_2$ along $A_1D_1H_2$ and $A_2D_2H_2$. Thus, these two heteroclinic orbits should satisfy the following conditions:

$$\lim_{t\to+\infty} x(t) = H_1, \qquad \lim_{t\to-\infty} x(t) = H_2, \tag{103}$$

First, we study the heteroclinic orbit $A_1B_1H_1$ and $A_1D_1H_2$. Let $L_0 = M_0 = 13$, the zero-order approximate solution $x_0$ can be obtained from Step I.

$$\begin{aligned}\mathrm{GPA}[13/13]_{x_0}(t) &= \Big(0.6454\mathrm{Tanh}(t) - 3.1816\mathrm{Tanh}(t)^3 \\ &\quad + 4.9227\mathrm{Tanh}(t)^5 - 3.2205\mathrm{Tanh}(t)^7 \\ &\quad + 0.9027\mathrm{Tanh}(t)^9 - 0.0879\mathrm{Tanh}(t)^{11} \\ &\quad + 0.0012\mathrm{Tanh}(t)^{13}\Big) \Big/ \Big(1 - 4.9299\mathrm{Tanh}(t)^2 \\ &\quad + 7.6444\mathrm{Tanh}(t)^4 - 5.06592\mathrm{Tanh}(t)^6 \\ &\quad + 1.4965\mathrm{Tanh}(t)^8 - 0.1833\mathrm{Tanh}(t)^{10} \\ &\quad + 0.0090\mathrm{Tanh}(t)^{12}\Big)\end{aligned} \tag{104}$$

Let $L_1 = M_1 = 11$, the first order approximate solution $x_1$ can be obtained from Step II.



Figure 3: Homoclinic orbits of the SD oscillator at large perturbation parameters ($\varepsilon = 50$, 60).

### 3.4 Simultaneous Homo-Heteroclinic Prediction

For oscillators where both homoclinic and heteroclinic orbits coexist at the same saddle (e.g., certain parameter regions of the DH-VDP oscillator), GPLP predicts **both critical parameter values simultaneously** (Fig. 12), a capability not available from standard perturbation methods.

## 4 Key Findings

1. **No-integration perturbation method:** GPLP eliminates the derivation/integration bottleneck of classical L–P, enabling high-order solutions for complicated oscillators.

2. **Multiple approximant types:** Different generalized Padé approximants are tailored to symmetric homoclinic, asymmetric heteroclinic, and mixed-mode orbits.

3. **Wide applicability:** Validated on Helmholtz–Duffing (polynomial), Duffing-harmonic (rational), and SD (irrational) oscillators.

4. **Simultaneous bifurcation prediction:** For the first time, both homoclinic and heteroclinic bifurcation parameters are predicted analytically in a unified framework.

Author: Zhenbo Li (lizhenbo@usc.edu.cn)
University of South China, School of Mathematics and Physics